Questions

Q1.

(i) The chart shows the electromagnetic (EM) spectrum.
Some parts of the spectrum have been labelled.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays

State the name of part **Q**.

(1)

.....

(ii) Three telescopes using different parts of the spectrum have been added to the chart.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays
Jodrell Bank			Hubble			Compton

Jodrell Bank is located near Manchester, whereas Hubble and Compton are in space.

Explain why some telescopes are located outside the Earth's atmosphere.

(2)

.....

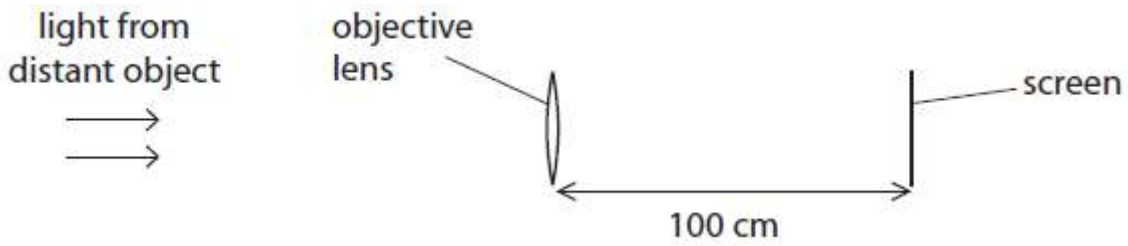
.....

Q2.

(a) This picture shows Galileo holding a telescope.

A student starts to make a similar telescope.

She places the objective lens near a screen to form a sharp image of a distant object.



(i) Use words from the box to complete the sentence.

(2)

bigger than the object	the right way up	the same size as the object
smaller than the object	upside down	

The sharp image formed is

..... and .....

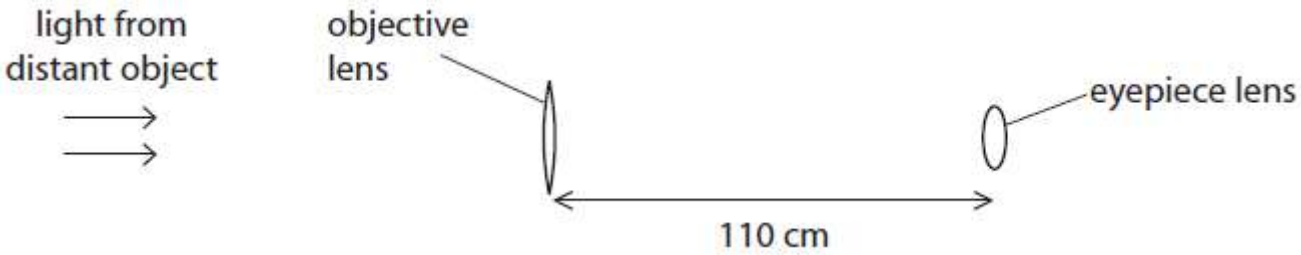
(ii) State how you can tell that the image formed is real.

(1)

.....

.....

(iii) She then removes the screen and replaces it with an eyepiece lens.
She moves the eyepiece lens until she can see a sharp, magnified image through it.
The diagram shows the position of the lenses.



The lenses are now separated by 110 cm.
Complete the sentence by putting a cross (☒) in the box next to your answer.
The focal length of the eyepiece is about

(1)

☐ **A** 10 cm
☐ **B** 55 cm
☐ **C** 90 cm
☐ **D** 100 cm

(iv) Complete the sentence by putting a cross (☒) in the box next to your answer.
At the eyepiece of this telescope there is

(1)

☐ **A** refraction by a converging lens
☐ **B** refraction by a converging mirror
☐ **C** reflection by a converging lens
☐ **D** reflection by a converging mirror

(b) State how the invention of the telescope improved the way scientists observed the Universe.

(1)

.....

.....

(c) The invention of the telescope helped to change ideas about the Solar System.
The heliocentric model replaced the geocentric model.
Describe the difference between the geocentric model and the heliocentric model.

(2)

.....

.....

.....

.....

(Total for Question = 8 marks)

Q3.

(i) Which type of surface will produce a specular reflection?

(1)

☐ **A** calm water

☐ **B** clothing

☐ **C** paper

☐ **D** road surface

(ii) A surface appears bright white when white light is shone onto it.

(1)

This is because it

☐ **A** absorbs all the white light

☐ **B** absorbs only red and blue light

☐ **C** absorbs only green and blue light

☐ **D** absorbs none of the white light

(Total for question = 2 marks)

Q4.

Figure 9 shows a metal cube filled with hot water.

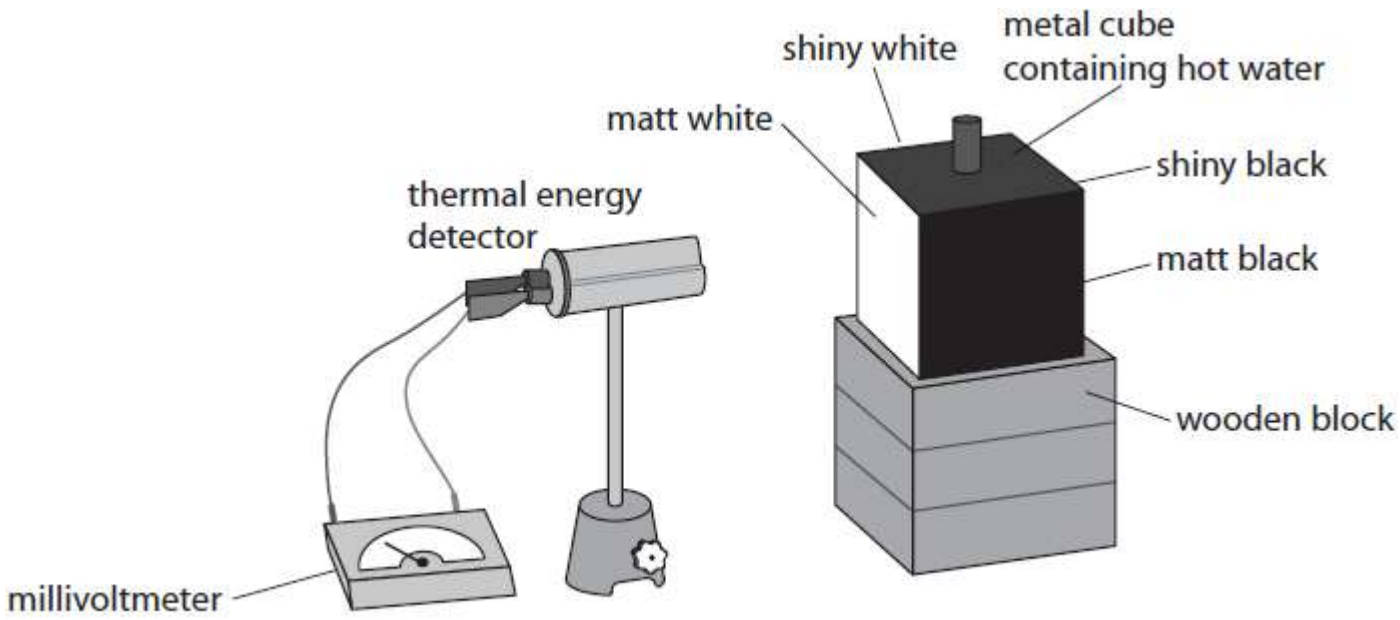


Figure 9

Four sides of the cube have different surfaces, as shown in Figure 9.

The four surfaces are at the same temperature.

The thermal energy radiated by each side of the box is measured using a thermal energy detector connected to a millivoltmeter.

The detector is moved to get a reading of the thermal energy emitted from each side of the box.

Figure 10 shows the table of results.

type of surface	millivoltmeter reading in mV
matt white	32
shiny white	20
shiny black	
matt black	55

Figure 10

(i) Suggest a possible millivoltmeter reading for the shiny black surface.

(1)

..... mV

(ii) State what must be kept the same to take the measurement for each surface.

(1)

.....

.....

(iii) Suggest why the cube is placed on a block of wood.

(1)

.....

.....

(Total for question = 3 marks)

Q5.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

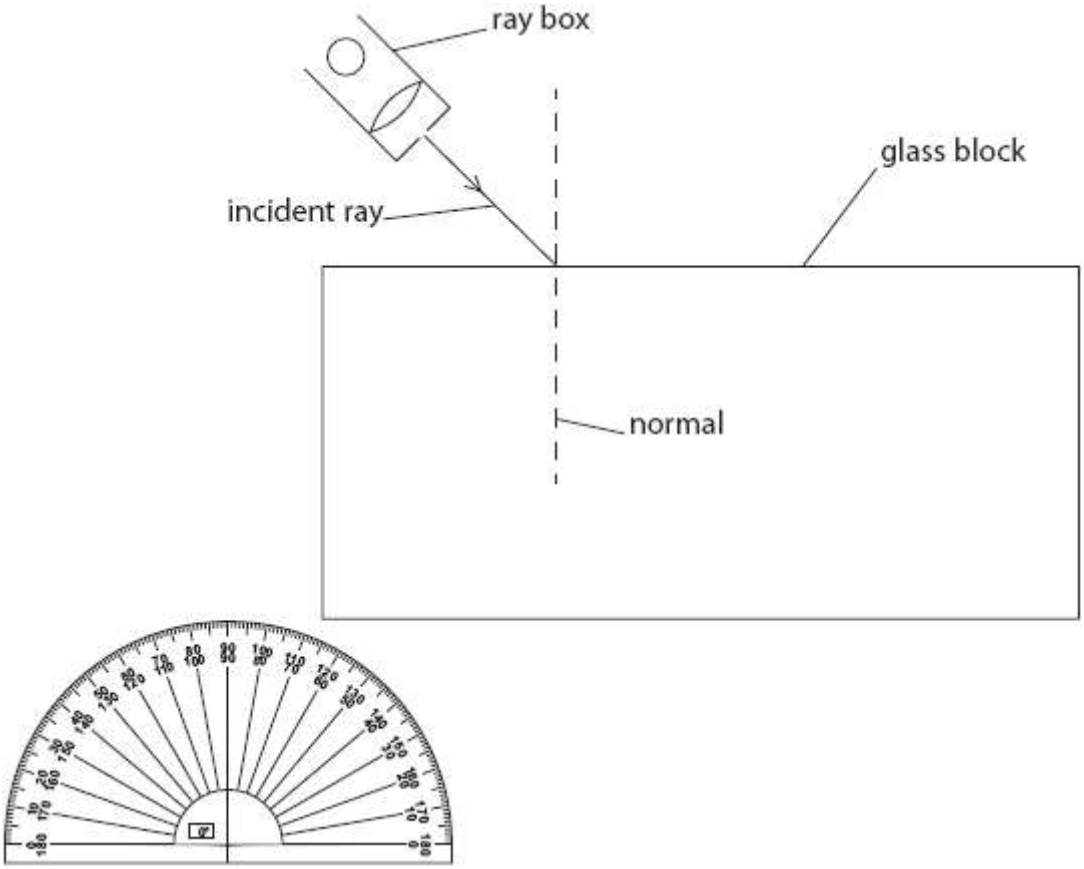


Figure 5

The student's results are shown in Figure 6.

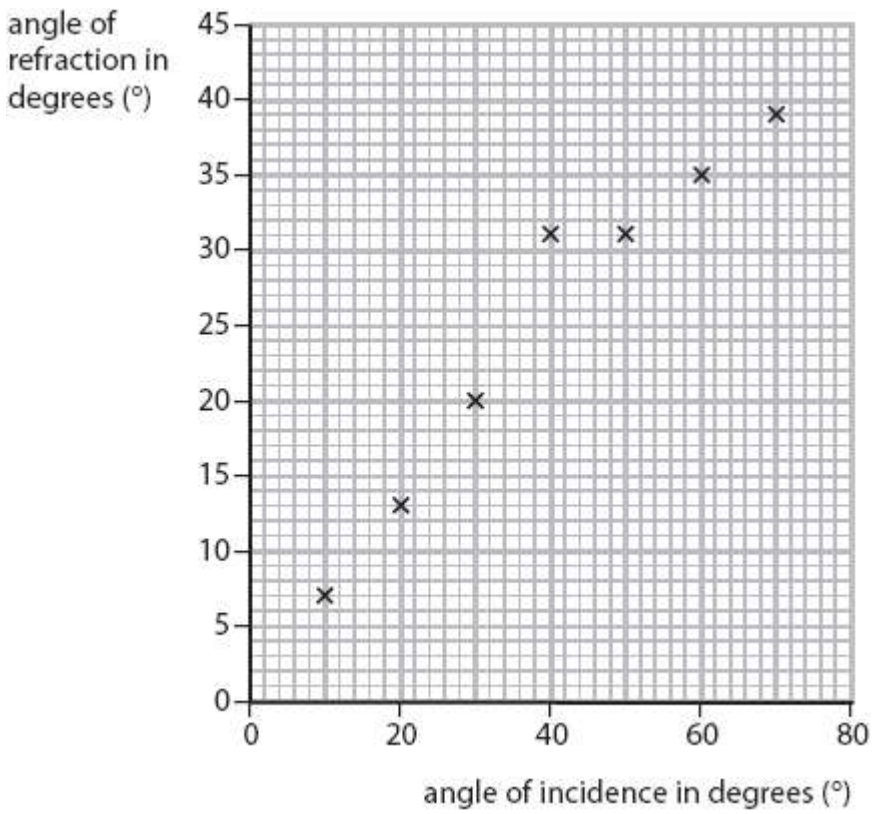


Figure 6

Explain whether any of the readings are anomalous.

(2)

.....

.....

.....

.....

(Total for question = 2 marks)

Q6.

This question is about the electromagnetic spectrum.

Figure 1 shows the full electromagnetic spectrum.

Figure 2 gives information about four of the types of electromagnetic radiation shown in Figure 1.

radio waves	microwaves	infrared	visible light	ultraviolet	x-rays	gamma rays

Figure 1

State the name of each type of radiation next to its information.

(4)

Information	Type of electromagnetic radiation
emitted by bright objects **and** can be detected by the human eye	.....
emitted by radioactive nuclei **and** can be used to treat cancer	.....
produced in an aerial **and** can be used for communication	.....
can cause skin cancer in humans **and** can be used to detect forged banknotes	.....

Figure 2

(Total for question = 4 marks)

Q7.

A flag has three colours in daylight.

The flag is taken into a dark room.

Figure 3 shows the flag when a torch shines white light on to the flag.

Figure 4 shows the same flag when the torch is covered with a coloured filter.

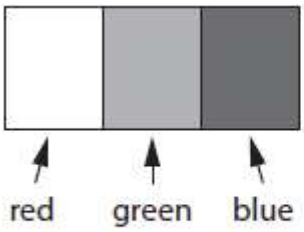


Figure 3

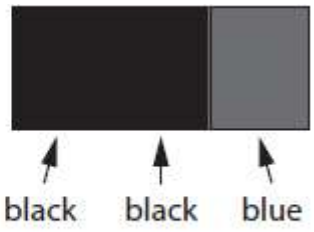


Figure 4

State the colour of the filter that has covered the torch.

(1)

.....

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 4 shows two lenses, P and Q, arranged to form a simple telescope.

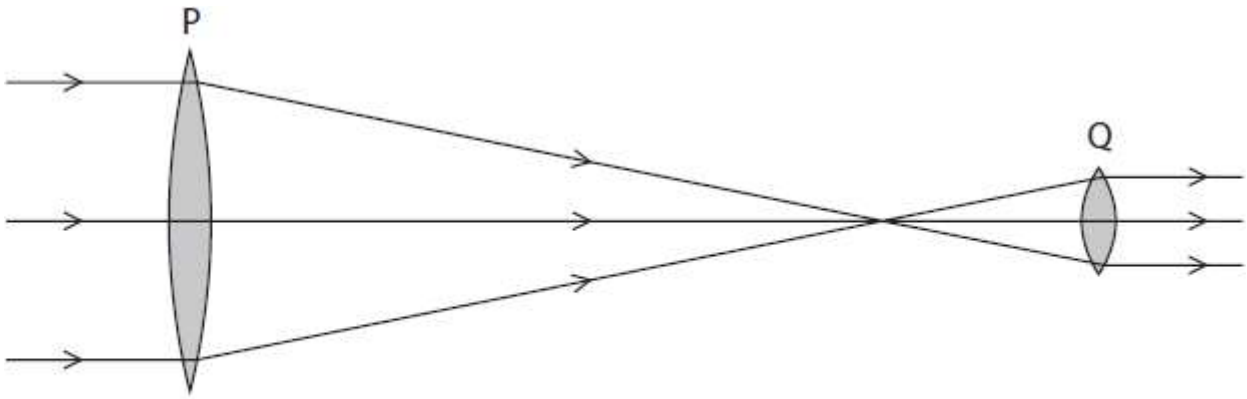


Figure 4

(i) State **one** use for a telescope.

(1)

.....

.....

(ii) Which of these describes the lenses?

(1)

☐ **A** P is converging, Q is diverging
☐ **B** P is diverging, Q is converging
☐ **C** P and Q are both converging
☐ **D** P and Q are both diverging

(iii) The focal length of lens Q is 0.14 m.

Calculate the power of the lens.

(2)

Use the equation

$$\text{power} = \frac{1}{\text{focal length in m}}$$

power of lens = ... dioptre

(Total for question = 5 marks)

Q9.

This question is about light.

Figure 9a and Figure 9b show rays of light before and after passing through different lenses.

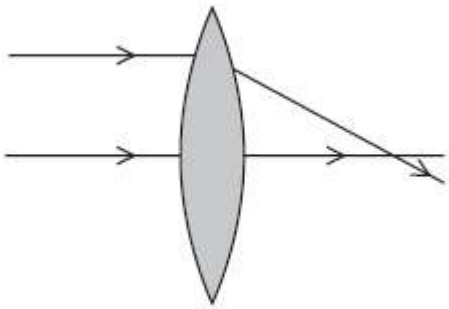


Figure 9a

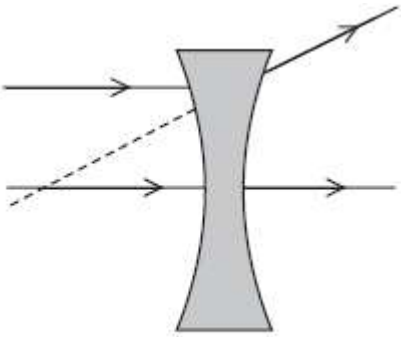


Figure 9b

(i) State **one** similarity and **one** difference in the way the rays of light pass through the lenses.

(2)

similarity

.....

.....

difference

.....

.....

(ii) Figure 10 shows two glass lenses P and Q.

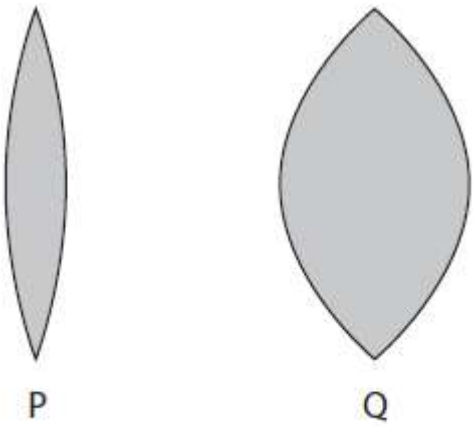


Figure 10

State how the power of P is different from the power of Q.

(1)

.....

.....

(Total for question = 3 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about light.

Figure 8 shows a shiny metal ball.

(Source: © frerd/Shutterstock)

Figure 8

A clear image of a building can be seen on the surface of the ball.

(i) This clear image is an example of

(1)

☐ **A** diffuse reflection
☐ **B** diffuse refraction
☐ **C** specular reflection
☐ **D** specular refraction

(ii) Explain why the surface of the metal ball gives a **clear** image.
You may draw diagrams to help with your answer.

(2)

.....

.....

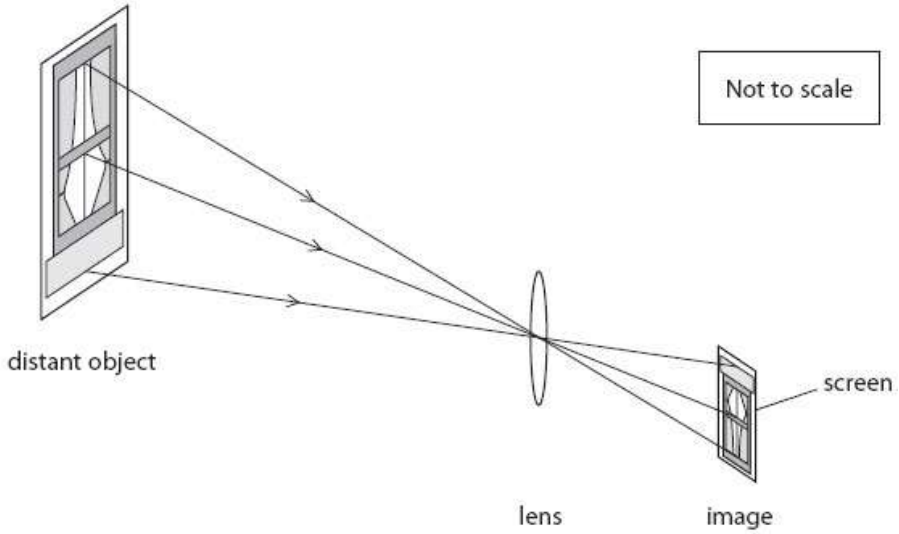
.....

.....

(Total for question = 3 marks)

Q11.

A lens can be used to produce a clear image of a distant object on a screen.



(a) (i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The image produced is real because it is

(1)

☐ **A** in focus

☐ **B** magnified

☐ **C** on a screen

☐ **D** smaller

(ii) Describe how to measure the focal length of the lens.

(2)

.....

.....

.....

.....

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
i	infrared (radiation)/(rays) (1)		**(1)**
ii	An explanation linking any two from • above the clouds / no clouds/ no weather (1) • image is clearer/more detailed/ not distorted/not blurred (1) • no light pollution (1) • (some) telescopes use gamma/ X-rays/ultraviolet /infrared/microwaves (1) • no absorption (by atmosphere) of gamma/ Xrays/ ultraviolet /infrared/ microwaves (1)	Credit to be given for stating that all telescopes would be better in space, but size and weight may exclude e.g. Jodrell Bank from space. no air/dust/pollution wider field of view/ can use anytime IG NORE 'see further' IGNORE 'it is closer (to the stars/planets)' IGNORE: references to improving understanding / knowledge of space	**(2)**

Q2.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	In ANY order smaller than the object (1) upside down (1)		**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	The idea that it can be captured on a screen	appears where the rays (of light) meet ignore references to reflection	**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	A (10)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iv)	A (refraction by a converging lens)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	image was larger/magnified (compared with naked eye).	better {information / detail} / see {further / clearer /further away} / fainter /more objects / zoom in OWTTE	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	Description to include: Geo = Earth at the centre (of the Solar System) (1) Helio = Sun at the centre (of the Solar System) (1)	"everything goes around the Earth / us" the Sun goes around the Earth "everything goes around the Sun" we revolve around the Sun Allow answers in diagram form Allow 1 mark for **both** descriptions the wrong way round	**(2)**

Q3.

Question number	Answer	Mark
(i)	A calm water	**(1)**

Question number	Answer	Mark
(ii)	D absorbs none of the white light	**(1)**

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	any value between 21 and 54 inclusive		**(1)** **AO3**

Question Number	Answer	Additional guidance	Mark
(ii)	any one from distance / separation (1) time (exposure) (1)		(1) AO3

Question Number	Answer	Additional guidance	Mark
(iii)	Any one from poor conductor / reduces energy loss / (good) insulator (1) same height as detector (1)	allow heat loss for energy loss accept wood stops heat loss accept it does not conduct heat	(1) AO3

Q5.

Question number	Answer	Additional guidance	Marks
	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification (1 mark):		(2)
	• anomalous result / it is at (angle of incidence) 40° (1)	the 4th cross the one in the middle. may be identified on the graph	
	• (because) it is further away from the curve of best fit than the other results (1)	does not follow same pattern as the others / not in line etc. explains the cause of the error e.g. misreading protractor suggests a plausible correct result such as 40(°),25(°) ignore points at i = 40(°) and 50(°) are the same	

Q6.

Question number	Answer	Additional guidance	Mark
	(visible) light (1) gamma (rays) (1) radio (waves) (1) ultraviolet (waves) (1)	 γ (rays) allow microwaves UV (rays)	**4** **AO3.3**

Q7.

Question number	Answer	Additional guidance	Mark
	blue		**1** **AO3.2**

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	Any one from:- to view distant objects /things far away (1) looking at planets /stars / space/ galaxies (1) bird watching (1)	 accept magnify distant objects accept other named astronomical objects	**(1)** **AO2**

Question Number	Answer	Mark
(ii)	**C** P and Q are both converging **A** is incorrect the lens Q is not diverging **B** is incorrect the lens P is not diverging **D** is incorrect neither lens is diverging	**(1)** **AO3**

Question Number	Answer	Additional guidance	Mark
(iii)	substitution (1) (power =) $\frac{1}{0.14}$ evaluation (1) 7.1 (D)	 7.14(28...) allow 7 award 1 mark for 7.1 to any other power of ten award 1 mark for correct answer as a fraction award full marks for the correct answer with no working	(2) AO2

Q9.

Question number	Answer	Additional guidance	Mark
(i)	similarity (both) change direction /bend/refract (rays of light) (1) OR (rays of light/they) pass/go (straight) through the (optical) centre / focus(1) difference one converges the other diverges (1)	 accept ray through centre described as 'bottom ray' accept 'top ray' accept refracts/bends in different ways do not allow 'change in direction of top ray' this is a similarity	(2) AO3

Question number	Answer	Additional guidance	Mark
(ii)	(the power of) P is less than (the power of) Q	ORA allow Q is greater /bigger	(1) AO2

Q10.

Question number	Answer	Additional guidance	Mark
(i)	C specular reflection A is incorrect as the reflection is not diffuse B is incorrect as it is not refraction D is incorrect as it is not refraction		(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	An explanation linking: (the surface/metal ball) is smooth/shiny (1) (for each ray of light) the angle of incidence is equal to the angle of reflection (1)	 like a mirror the reflection is even / there is no scattering full marks can be awarded for labelled diagrams	(2) AO2

Q11.

	Answer	Acceptable answers	Mark
(i)	☒ C on a screen		**(1)**
(ii)	Explanation linking the following:- (measure) distance / length / from / line (1) <u>lens</u> to image / screen / focal point (1)	how far away use ruler / tape measure measure u(object distance) and v(image distance) calculate focal length using 1/u + 1/v = 1/f (measure) from lens to image (2)	**(2)**